EX27 · CLASS_17_VECTORS · 01 / ENGINEERING AND MATH THEORY

Cross product and moments

Vector algebra in R³, geometric meaning and force moments.

Learning sequence
01 Math theory → 02 Python reference → 03 Exercise → 04 Pseudocode → Programming.

Prerequisites: EX25 and EX26, NumPy arrays, Euclidean norm and dot product.

1. The mathematical idea

In EX26, the dot product combined two vectors and returned a **scalar**. The cross product also combines two vectors, but yields a **vector perpendicular to both**. Its magnitude depends on the input lengths and the angle between them. We study it mathematically first.

2. Notation and three-dimensional vectors

We work with three-component vectors in Euclidean R³. Vectors are bold, while their coordinates are numbers. Hatted letters represent the Cartesian unit vectors, **not the imaginary unit**.

$$\mathbf{u} = (u_x, u_y, u_z), \quad \mathbf{v} = (v_x, v_y, v_z)$$

$$\mathbf{u} = u_x\hat{\imath} + u_y\hat{\jmath} + u_z\hat{k}$$

The basis directions satisfy:

$$\hat{\imath} \times \hat{\jmath} = \hat{k}, \quad \hat{\jmath} \times \hat{\imath} = -\hat{k}$$

3. Coordinate definition

The operation is written × (it is neither scalar multiplication nor Python `*`). In a right-handed Cartesian basis its coordinate definition is:

$$\mathbf{u} \times \mathbf{v} = (u_y v_z - u_z v_y,\ u_z v_x - u_x v_z,\ u_x v_y - u_y v_x)$$

The same expression can be written as a linear combination of Cartesian unit vectors. Its sign pattern follows the first-row expansion of the symbolic determinant:

$$\mathbf{u} \times \mathbf{v} = (u_y v_z - u_z v_y)\hat{\imath} + (u_z v_x - u_x v_z)\hat{\jmath} + (u_x v_y - u_y v_x)\hat{k}$$

4. Core algebraic properties

Property	Meaning
Anti-commutativity	u × v = −(v × u). Order matters.
Distributivity	u × (v + w) = u × v + u × w
Homogeneity	(c u) × v = c(u × v)
Perpendicularity	(u × v) · u = (u × v) · v = 0
Parallel vectors	u × v = 0; also true if either input is zero.

Warning: the cross product is **not associative** in general. (u × v) × w need not equal u × (v × w).

5. Geometric meaning: magnitude and orientation

$$\|\mathbf{u} \times \mathbf{v}\|_2 = \|\mathbf{u}\|_2 \|\mathbf{v}\|_2 \sin\theta$$

This magnitude equals the **area of the parallelogram** spanned by the vectors. For perpendicular vectors sine equals 1; for parallel vectors it is zero. The **right-hand rule** determines the result direction in a right-handed coordinate system.

> **Important convention**
> +z corresponds to turning from +x toward +y by the right-hand rule. Reversing the vector order changes the sign.

6. Step-by-step algebraic example

Given: two vectors without physical units.

$$\mathbf{u} = (2, -1, 3), \quad \mathbf{v} = (1, 4, -2)$$

First component:

$$(\mathbf{u} \times \mathbf{v})_x = (-1)(-2) - (3)(4) = 2 - 12 = -10$$

Second component:

$$(\mathbf{u} \times \mathbf{v})_y = (3)(1) - (2)(-2) = 3 + 4 = 7$$

Third component:

$$(\mathbf{u} \times \mathbf{v})_z = (2)(4) - (-1)(1) = 8 + 1 = 9$$

Vector result:

$$\mathbf{u} \times \mathbf{v} = (-10, 7, 9)$$

Euclidean norm:

$$\|\mathbf{u}\times\mathbf{v}\|_2=\sqrt{(-10)^2+7^2+9^2}=\sqrt{230}\approx 15.166$$

Perpendicularity check:

$$\mathbf{u}\cdot(\mathbf{u}\times\mathbf{v})=2(-10)+(-1)(7)+3(9)=0$$
$$\mathbf{v}\cdot(\mathbf{u}\times\mathbf{v})=1(-10)+4(7)+(-2)(9)=0$$

These checks validate the result **before programming**. We will later compare with NumPy.

7. Physics application: moment of a force

Suppose a force acts at point P and we want its moment about O. The position vector **r** must point from O to P. By definition:

$$\mathbf{M}_O=\mathbf{r}_{OP}\times\mathbf{F}$$

The moment is a vector measured in N·m. Its magnitude expresses a turning effect about the chosen point. Although N·m and joules share SI base units, torque/moment and energy are different physical quantities.

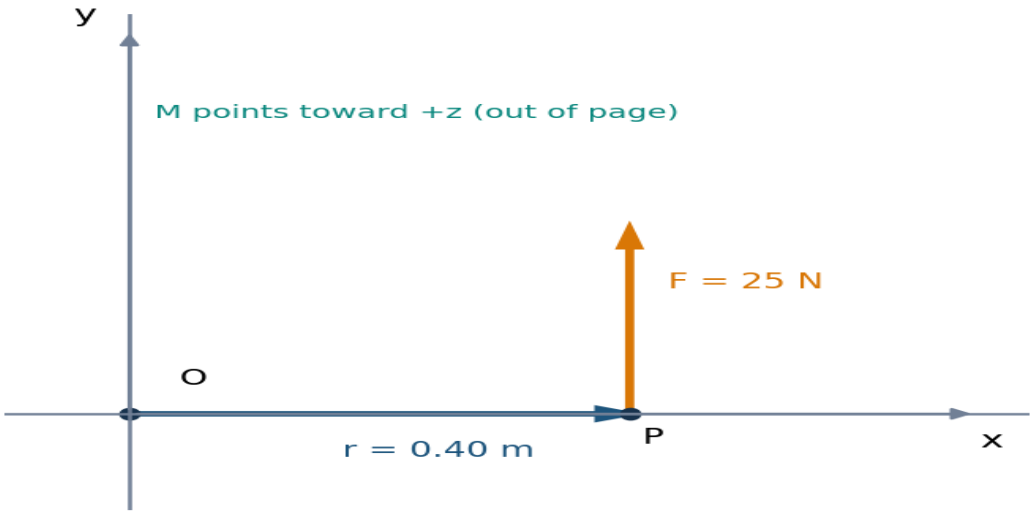


Diagram: the lever arm is along +x and the force along +y. The resulting moment points along +z.

Given:

$$\mathbf{r}=(0.40,0,0)\text{ m},\quad \mathbf{F}=(0,25,0)\text{ N}$$

Cross product:

$$\mathbf{M}_O=(0,0,\ 0.40\cdot 25)=(0,0,10)\text{ N m}$$

Interpretation: the moment magnitude is 10 N·m in the +z direction. Reversing the arm or force reverses the moment sign.

8. Planar problems and 3-component vectors

Even when force and lever arm lie in the xy-plane, the moment is normal to that plane. For `np.cross`, we represent planar vectors as (x,y,0), explicitly keeping the z coordinate.

$$\mathbf{r} = (x, y, 0),\ \mathbf{F} = (F_x, F_y, 0) \implies \mathbf{M} = (0, 0, xF_y - yF_x)$$

9. Python connection and takeaways

In Python, `np.cross(u, v)` computes the cross product; `np.dot()` checks perpendicularity; `np.linalg.norm()` gives the magnitude. Check dimensions first; interpret signs and units afterwards.

Remember
Dot product → scalar and angle. Cross product → normal vector and area. Moment → lever-arm cross force, in the order r × F.